

Producer Theory

Lecture 16: Short Run & Long Run Market Equilibrium

Paulo Fagandini

2026

Recap: Where We Are

Over Lectures 5–15, we built **both sides** of the market:

Demand side (L5–L9):

- Utility maximization → demand curve
- Market demand (horizontal sum)
- Consumer surplus
- Demand elasticity

...

 **Today:** We put them **together** to find the market equilibrium — the price and quantity where buyers and sellers agree. This is the **payoff** of the entire course so far!

Supply side (L10–L15):

- Cost structure → $P = MC$ → supply curve
- Market supply (horizontal sum)
- Producer surplus
- Supply elasticity

Short-Run Market Equilibrium

Where Supply Meets Demand

MARKET EQUILIBRIUM

The price and quantity at which **quantity demanded equals quantity supplied**:

$$Q_d(P^*) = Q_s(P^*)$$

At the equilibrium price P^* , every buyer who wants to buy at that price can, and every seller who wants to sell can. **No one is left unsatisfied.**

Finding equilibrium algebraically (linear curves):

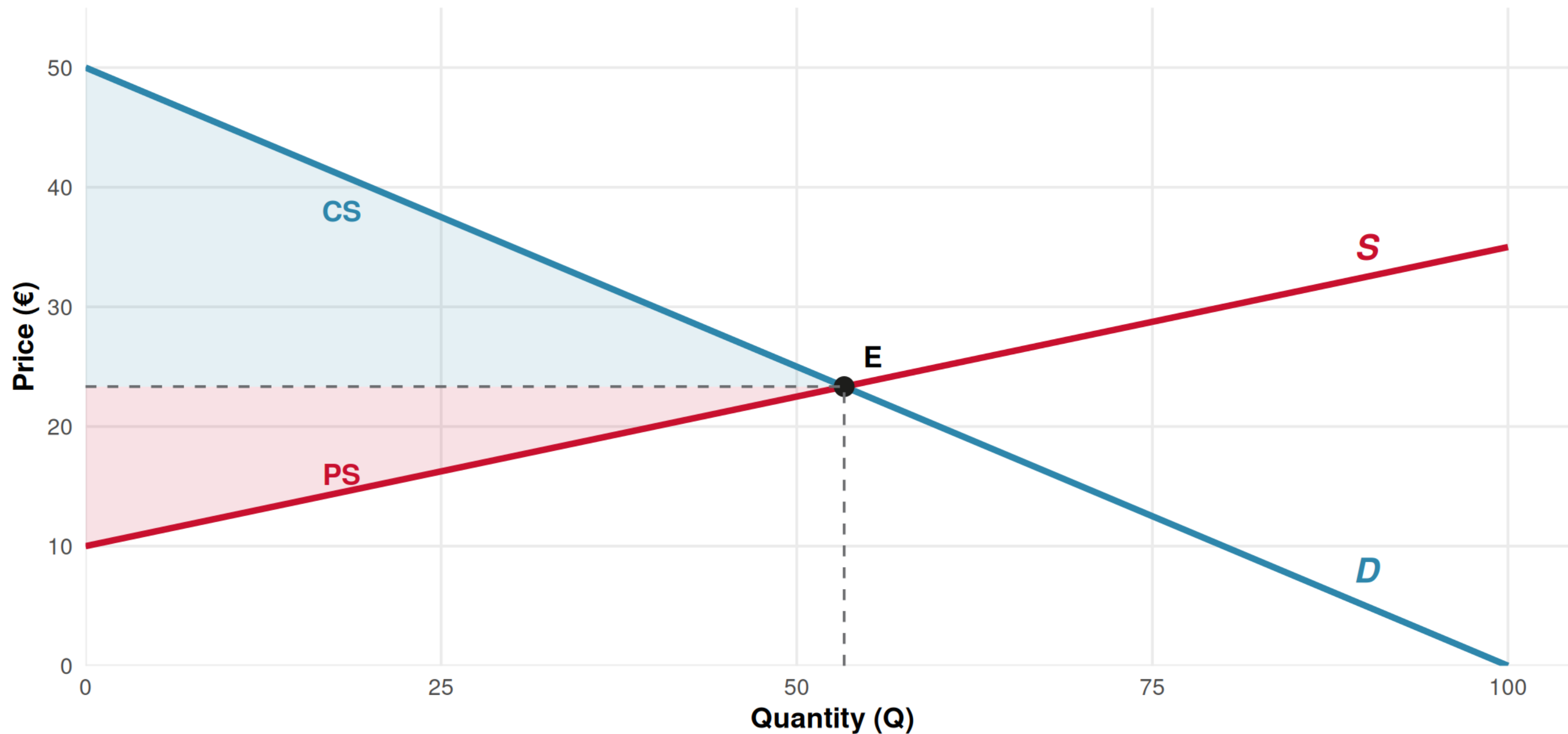
Given $D : P = 50 - 0.5Q$ and $S : P = 10 + 0.25Q$

Set equal: $50 - 0.5Q = 10 + 0.25Q \rightarrow 40 = 0.75Q \rightarrow Q^* = 53.3 \rightarrow P^* = €23.3$

 At this price, the quantity buyers want to buy **exactly matches** the quantity sellers want to sell.

Equilibrium Graphically

Market Equilibrium with CS and PS (Hypothetical)



What Happens Away from Equilibrium?

Price too HIGH ($P > P^*$): **Surplus** (excess supply)

- $Q_s > Q_d \rightarrow$ goods pile up
- Sellers compete by **lowering prices**
- Price falls toward P^*

 *Tourism*: Hotels with too many empty rooms in off-season
→ prices drop on Booking.com

Price too LOW ($P < P^*$): **Shortage** (excess demand)

- $Q_d > Q_s \rightarrow$ not enough goods
- Buyers compete by **bidding prices up**
- Price rises toward P^*

 *Tourism*: Flight demand exceeds seats during holidays →
prices rise, last-minute fares skyrocket

...

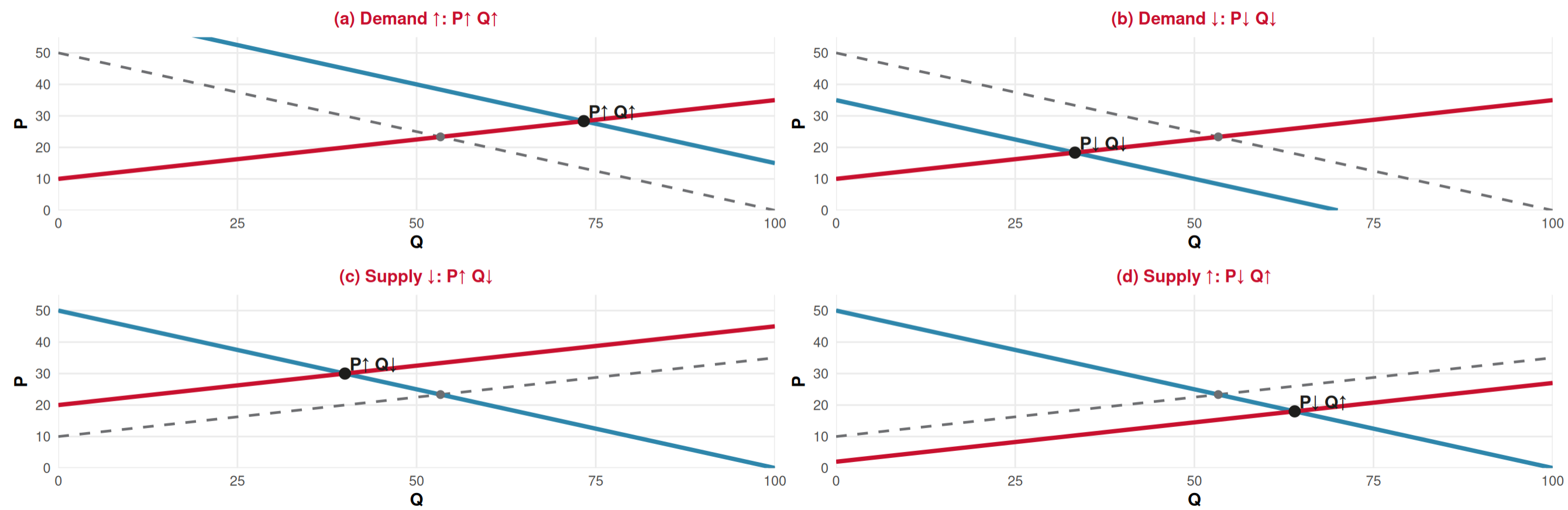
SELF-CORRECTING MECHANISM

Markets **naturally tend toward equilibrium**. Surpluses push prices down; shortages push prices up. This process continues until $Q_d = Q_s$.

The Four Shift Rules



The Four Shift Rules



The Four Rules Summarized

Shift	Price	Quantity	Tourism Example
Demand ↑ (right)	↑	↑	Summer season → more tourists want hotels
Demand ↓ (left)	↓	↓	Pandemic → fewer tourists travel
Supply ↑ (right)	↓	↑	New technology → cheaper flights
Supply ↓ (left)	↑	↓	Fuel crisis → airline costs rise

...

 **When both curves shift simultaneously**, the direction of **one** variable (P or Q) becomes **ambiguous** — it depends on the **relative size** of the shifts.

Example: Summer in the Algarve — demand shifts right (tourists) AND supply shifts right (seasonal workers). Quantity definitely rises, but price? Depends on which shift is bigger!

Welfare at Equilibrium

TOTAL ECONOMIC WELFARE

$$W = CS + PS$$

At the competitive equilibrium, total welfare is **maximized**. Any other price/quantity combination would reduce the combined surplus.

From our example ($D : P = 50 - 0.5Q$, $S : P = 10 + 0.25Q$, $Q^* \approx 53.3$, $P^* \approx 23.3$):

$$CS = \frac{1}{2} \times 53.3 \times (50 - 23.3) = \frac{1}{2} \times 53.3 \times 26.7 \approx \text{€}711$$

$$PS = \frac{1}{2} \times 53.3 \times (23.3 - 10) = \frac{1}{2} \times 53.3 \times 13.3 \approx \text{€}356$$

$$W = 711 + 356 \approx \text{€}1,067$$

👉 Any **tax**, **price ceiling**, or **price floor** would reduce total welfare by creating **deadweight loss**.

Long-Run Market Equilibrium

The Short Run vs Long Run Revisited

Short-run equilibrium:

- Number of firms is **fixed**
- Firms may earn **positive profit**, **zero profit**, or **losses**
- Some factors of production are fixed
- This is what we've been analyzing!

...

Long-run equilibrium:

- Firms can **enter** or **exit** the market
- All factors of production are variable
- A special condition emerges: **zero economic profit**
- The market adjusts to a new, more fundamental equilibrium

THE KEY LONG-RUN QUESTION

What happens over time when firms in a competitive market are earning positive profits? Or making losses?

The Entry and Exit Mechanism

If firms earn positive economic profit ($P > ATC$):

1.  New firms are attracted → they **enter** the market
2.  Market supply shifts **right** (more firms = more supply)
3.  Equilibrium price **falls**
4.  Process continues until $P = ATC_{\min} \rightarrow$ **zero economic profit**

If firms make losses ($P < ATC$):

1.  Some firms **exit** the market
2.  Market supply shifts **left** (fewer firms = less supply)
3.  Equilibrium price **rises**
4.  Process continues until $P = ATC_{\min} \rightarrow$ **zero economic profit**

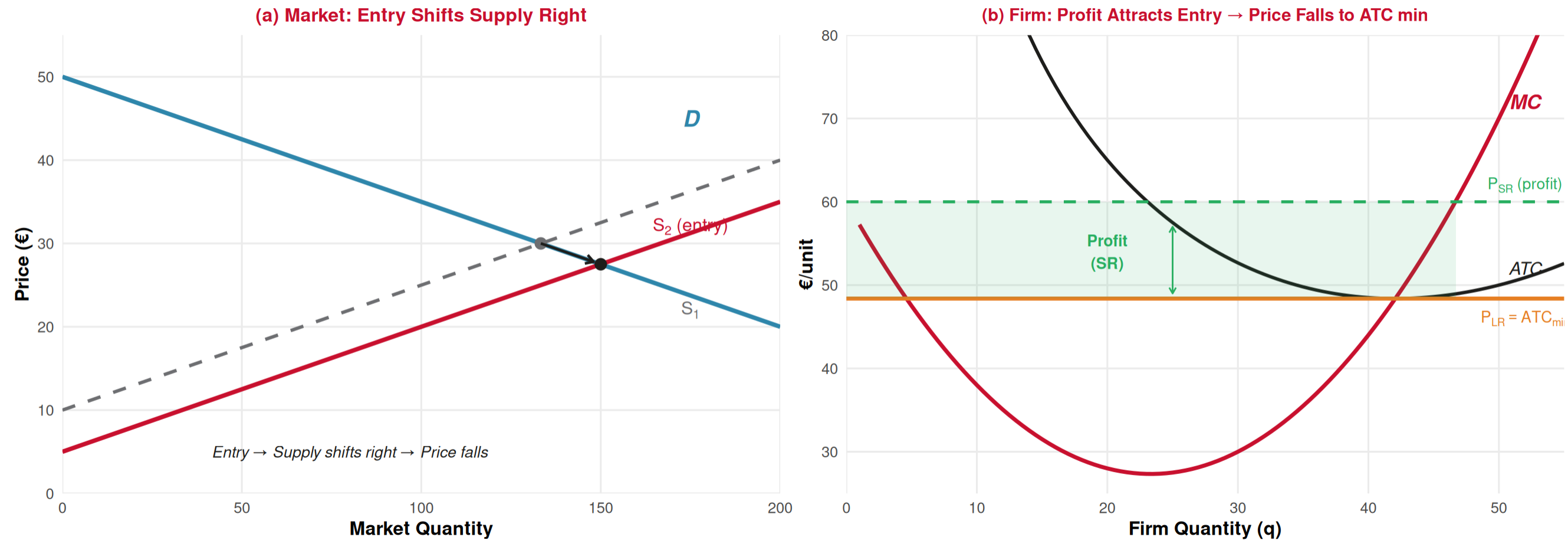
...

LONG-RUN EQUILIBRIUM CONDITION

In a perfectly competitive market: $P = MC = ATC_{\min}$

Firms produce at minimum average cost and earn **zero economic profit**.

Visualizing the Long-Run Adjustment



(a) Positive profits attract **entry** → supply shifts right → price falls. **(b)** Price falls until $P = ATC_{min}$ → zero economic profit → **entry stops**.

Zero Profit $\neq$ Failure!

“Zero economic profit” sounds bad — but remember what it means:

ZERO ECONOMIC PROFIT

The firm covers **all** costs, including the **opportunity cost** of the owner’s time and capital. The owner earns exactly what they could earn in their **best alternative** — a **normal return**.

$$\text{Economic Profit} = 0 \implies \text{Accounting Profit} = \text{Normal Return}$$

✓ What zero economic profit means:

- The business is sustainable
- The owner has no reason to leave
- No new firms want to enter
- The market is in **long-run rest**

✗ What it does NOT mean:

- The owner earns no money
- The business is failing
- The owner should close
- There are no accounting profits

The Long-Run Supply Curve

If all firms have the **same cost structure** (constant-cost industry):

LONG-RUN SUPPLY IN A CONSTANT-COST INDUSTRY

The long-run supply curve is **horizontal** at $P = ATC_{\min}$.

- If demand increases → new firms enter, supply shifts right, price returns to $ATC_{\min}$
- If demand decreases → firms exit, supply shifts left, price returns to $ATC_{\min}$
- **Quantity adjusts**, but **price stays the same** in the long run!

👉 In reality, costs may **rise** as the industry grows (increasing-cost industry → upward-sloping LR supply) or **fall** (decreasing-cost industry → downward-sloping LR supply). But the constant-cost case is the baseline.

Tourism: The long-run supply of standardized tour packages may be nearly horizontal — if tour operators can easily enter/exit and all face similar costs, the long-run price settles at the minimum cost of providing a tour.

Tourism Application

Tourism Markets: SR vs LR

What happened to Lisbon's hotel market after the tourism boom (post-2015)?

1 Short run (2015–2017):

- Demand surged (low-cost airlines, Lisbon's popularity)
- Supply was **fixed** (existing hotels)
- Result: **prices skyrocketed**, occupancy hit record highs
- Existing hotels earned **large profits**
- $P \gg ATC_{\min} \rightarrow$ well above breakeven

2 Long run (2018–2024):

- Profits attracted **entry**: new hotels, Airbnb, hostels
- Supply shifted **right** significantly
- Result: prices stabilized (or even fell for some segments)
- Competition increased, profit margins **narrowed**
- Market moved toward $P \approx ATC_{\min}$

👉 This is the **entry mechanism** in action!

...



The pandemic (2020–2021) reversed this: demand collapsed → firms exited → supply shifted left → the cycle restarted.

Summary

Today's Key Takeaways:

1. **SR equilibrium**: where $Q_d = Q_s$. Price too high $\rightarrow$ surplus (price falls). Price too low $\rightarrow$ shortage (price rises).
2. **Four shift rules**: $D \uparrow \rightarrow P \uparrow Q \uparrow$; $D \downarrow \rightarrow P \downarrow Q \downarrow$; $S \uparrow \rightarrow P \downarrow Q \uparrow$; $S \downarrow \rightarrow P \uparrow Q \downarrow$
3. **Simultaneous shifts**: direction of one variable is ambiguous — depends on magnitude
4. **Total welfare** = CS + PS, maximized at competitive equilibrium
5. **LR equilibrium**: entry (if profit > 0) and exit (if profit < 0) drive $P \rightarrow ATC_{\min}$
6. **Zero economic profit** in the LR = normal return = sustainable. It's not failure!
7. **LR supply** (constant-cost): horizontal at $P = ATC_{\min}$

This completes the Producer Theory block! 🎉

Lectures 10–16: Costs $\rightarrow$ Profit max $\rightarrow$ Shutdown $\rightarrow$ Supply rule $\rightarrow$ PS & Market supply $\rightarrow$ Elasticity $\rightarrow$ Equilibrium

Next (Lecture 17, April 17): Introduction to Game Theory — Prisoner's Dilemma & Nash Equilibrium ♟️

Exercises

Practice Time! 

Market equilibrium, shifts, and long-run adjustment.

Exercise 1: Multiple Choice

Question: In the Algarve hotel market, a new low-cost airline begins flying to Faro from Northern Europe. At the same time, the Portuguese government introduces a new tourist tax on hotel stays. What is the effect on the equilibrium **quantity** of hotel room-nights?

- A. Quantity definitely increases
- B. Quantity definitely decreases
- C. Quantity definitely stays the same
- D. The effect on quantity is ambiguous

Answer: D

The new airline increases demand (shift right $\rightarrow Q\uparrow$). The tourist tax increases costs (supply shifts left $\rightarrow Q\downarrow$). Since one shift pushes Q up and the other pushes Q down, the net effect is **ambiguous** — it depends on which shift is larger. However, we can say that price **definitely rises** (both shifts push P upward).

Exercise 2: Multiple Choice

Question: In a perfectly competitive market, if firms are currently earning positive economic profits in the short run, what will happen in the long run?

- A. The government will impose a price ceiling
- B. New firms enter, supply shifts right, price falls until profit = 0
- C. Existing firms raise prices to increase profits further
- D. Demand shifts left as consumers find substitutes

Answer: B

Positive profit → **entry** → supply shifts right → equilibrium price falls → profit is competed away until $P = ATC_{\min} \rightarrow$ **zero economic profit**. This is the self-correcting mechanism of competitive markets. No government action needed (A), firms can't set prices (C — they're price takers!), and the question is about the supply side, not demand (D).

Exercise 3: Open Question

The market for guided walking tours in Sintra has the following curves:

- **Demand:** $P = 40 - 0.2Q$ (€ per tour)
- **Short-run supply:** $P = 8 + 0.12Q$ (€ per tour)

All firms are identical with $ATC_{\min} = €20$.

- Find the short-run equilibrium price and quantity.
- Calculate CS, PS, and total welfare at the SR equilibrium.
- Are firms making positive economic profit, zero profit, or losses? How do you know?
- Describe what will happen in the **long run**. What will the new equilibrium price be? Will the number of firms increase or decrease?
- A surge in tourist arrivals shifts demand to $P = 52 - 0.2Q$. Find the new SR equilibrium. Are firms now earning profits or losses?
- After the demand surge, describe the long-run adjustment process. What is the final LR equilibrium price?

Exercise 3: Solution — Parts a & b

a) SR equilibrium: $40 - 0.2Q = 8 + 0.12Q$

$$32 = 0.32Q \rightarrow Q^* = 100 \text{ tours}, P^* = 40 - 0.2(100) = \text{€}20$$

b) At $(Q^* = 100, P^* = \text{€}20)$:

$$CS = \frac{1}{2} \times 100 \times (40 - 20) = \frac{1}{2} \times 100 \times 20 = \text{€}1,000$$

$$PS = \frac{1}{2} \times 100 \times (20 - 8) = \frac{1}{2} \times 100 \times 12 = \text{€}600$$

$$W = CS + PS = 1,000 + 600 = \text{€}1,600$$

Exercise 3: Solution — Parts c & d

c) $P^* = €20 = ATC_{\min} = €20$

Since price equals minimum ATC, firms are earning **zero economic profit**. The market is already at its **long-run equilibrium!** 

d) Since profit is already zero, there is **no incentive** for firms to enter or exit. The long-run equilibrium price **stays at €20**. The number of firms remains the same.

👉 This is a special case where SR equilibrium **coincides** with LR equilibrium — the market is at rest.

Exercise 3: Solution — Parts e & f

e) New demand: $P = 52 - 0.2Q$

New SR equilibrium: $52 - 0.2Q = 8 + 0.12Q \rightarrow 44 = 0.32Q \rightarrow Q^* = 137.5, P^* = 52 - 0.2(137.5) = €24.50$

Since $P^* = €24.50 > ATC_{\min} = €20$: firms are now earning **positive economic profit**!

Profit per firm = $(24.50 - 20) \times q_i = €4.50 \times q_i > 0$ (exact firm quantity depends on individual MC).

f) Long-run adjustment:

1. Positive profits attract **new firms** → they enter the market
2. Supply shifts **right** (more firms)
3. Equilibrium price **falls** from €24.50 toward $ATC_{\min}$
4. Entry continues until $P = ATC_{\min} = €20$

Final LR equilibrium: $P_{LR} = €20$. At this price with the new demand: $Q = \frac{52-20}{0.2} = 160$ tours.

The market serves **more tours** (160 vs 100 originally) at the **same price** (€20). The adjustment came entirely through **more firms** entering, not through higher prices. 

Next Lecture

April 17, 2026: Introduction to Game Theory — Prisoner's Dilemma & Nash Equilibrium 

A brand new topic — what happens when firms are **not** price takers and **interact strategically**?

 **This completes the Producer Theory block!** Time to move beyond perfect competition.

Thank You!

Questions? 🙋

✉️ paulo.fagandini@ext.universidadeeuropeia.pt

Next class: Friday, April 17, 2026